

YAKUTSK, Russian Federation

WMO: 249590

Lat: 62.017N

Lon: 129.717E

Elev: 330

StdP: 14.52

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -54.2	(c) -51.2	(d) -58.2	(e) 0.2	(f) -53.9	(g) -55.3	(h) 0.2	(i) -51.1	(j) 10.7	(k) -26.1	(l) 9.5	(m) -27.1	(n) 1.6	(o) 340	(p) 0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	86.8	66.1	83.2	64.8	79.7	63.3	69.0	81.7	67.0	79.2	64.9	76.5	6.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	92.0	74.5	62.3	85.0	73.5	60.1	78.6	71.0	33.5	81.6	31.8	79.0	30.2	76.3	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.9	(b) 12.1	(c) 10.7	(d) DB	(e) -55.1	(f) 91.8	(g) 4.5	(h) 2.8	(i) -58.3	(j) 93.9	(k) -60.9	(l) 95.5	(m) -63.4	(n) 97.1	(o) -66.7	(p) 99.2	(q) 78.4
			(d) WB	(e) -55.1	(f) 72.5	(g) 4.5	(h) 2.3	(i) -58.3	(j) 74.1	(k) -60.9	(l) 75.5	(m) -63.4	(n) 76.7	(o) -66.7	(p) 78.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	-35.4	-27.2	-1.9	25.2	45.6	61.5	66.6	59.9	43.2	19.3	-15.1	-34.6		
	DBStd	38.52	9.81	10.19	12.98	10.37	7.81	7.41	6.80	6.54	7.52	11.77	14.35	10.30		
	HDD50	13118	2649	2163	1610	745	182	7	1	6	227	952	1953	2623		
	HDD65	17529	3114	2583	2075	1195	602	151	64	183	654	1417	2403	3088		
	CDD50	1248	0	0	0	0	45	351	515	314	23	0	0	0		
	CDD65	186	0	0	0	0	0	45	114	27	0	0	0	0		
	CDH74	2348	0	0	0	0	15	668	1322	340	3	0	0	0		
	CDH80	700	0	0	0	0	1	179	442	78	0	0	0	0		
(14) Wind	WSAvg	4.3	2.4	2.5	3.7	5.2	6.2	5.9	5.0	4.8	5.1	4.5	3.2	2.4		
(15) Precipitation	PrecAvg	9.2	0.4	0.3	0.2	0.4	0.7	1.3	1.6	1.5	1.1	0.7	0.6	0.4		
	PrecMax	12.8	0.9	0.9	0.8	0.9	2.3	3.4	3.9	5.9	3.4	1.9	1.7	0.8		
	PrecMin	5.3	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.1	0.1	0.2	0.2	0.1		
	PrecStd	1.8	0.2	0.2	0.2	0.2	0.6	0.7	0.9	1.1	0.7	0.4	0.3	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.8	11.9	37.9	56.3	76.2	88.7	92.6	86.7	70.7	45.8	28.4	-0.6		
		MCWB	-3.3	10.8	31.2	42.7	55.8	64.7	68.7	67.0	56.5	39.4	27.0	-1.2		
	2%	DB	-12.1	-0.3	30.2	50.0	70.1	85.0	88.3	81.6	64.6	40.6	19.3	-9.9		
		MCWB	-12.5	-1.0	25.4	39.1	52.7	63.7	67.3	65.2	52.6	35.2	17.8	-10.3		
	5%	DB	-16.9	-7.1	25.2	45.1	65.6	81.4	85.0	77.5	60.3	36.8	11.9	-15.4		
		MCWB	-17.2	-7.6	21.4	35.9	49.9	62.0	66.1	62.6	50.9	33.1	11.0	-15.7		
	10%	DB	-21.4	-12.6	19.5	41.1	61.2	77.7	81.6	73.8	56.3	33.8	5.2	-20.2		
		MCWB	-21.6	-13.0	16.6	33.6	47.7	60.2	65.2	61.0	48.8	30.8	4.5	-20.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-3.3	10.7	31.2	43.6	56.9	69.0	72.8	70.0	57.7	40.3	27.1	-1.5		
		MCDB	-2.9	11.7	37.5	55.0	73.1	83.5	85.8	81.3	66.3	45.1	28.2	-1.1		
	2%	WB	-12.4	-1.1	25.6	39.5	53.7	65.7	70.1	67.0	55.1	36.2	18.1	-10.5		
		MCDB	-12.0	-0.5	30.2	49.1	67.3	80.0	83.1	78.3	62.6	39.5	19.3	-10.1		
	5%	WB	-17.2	-7.6	21.5	36.5	51.0	63.6	68.4	64.6	52.4	33.3	11.0	-15.8		
		MCDB	-17.0	-7.1	25.2	44.2	63.2	77.5	81.0	75.2	58.5	36.3	11.8	-15.5		
	10%	WB	-21.6	-12.9	16.7	34.0	48.7	61.6	66.5	62.2	49.6	30.8	4.5	-20.4		
		MCDB	-21.4	-12.5	19.4	40.5	59.6	75.1	78.5	71.5	55.3	33.7	5.2	-20.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	15.4	24.2	22.0	20.3	21.8	20.8	20.2	17.3	13.0	12.2	8.8		
		MCDBR	16.0	21.3	26.7	22.7	28.1	28.9	27.8	27.5	24.9	14.9	17.1	14.9		
		MCWBR	15.8	20.8	22.5	13.9	14.3	12.6	12.3	13.7	15.1	10.9	16.5	14.7		
		MCDBR	16.1	21.4	26.3	21.6	24.5	25.1	23.4	24.1	21.0	14.3	17.0	15.0		
	5% WB	MCWBR	15.8	20.9	22.2	13.8	13.1	11.8	11.6	13.0	13.8	11.1	16.5	14.7		
(40) Clear-Sky Solar Irradiance	taub		0.213	0.257	0.299	0.357	0.425	0.419	0.495	0.468	0.362	0.287	0.219	0.183		
	taud		1.972	2.082	2.075	2.075	2.122	2.253	2.023	2.056	2.322	2.299	1.996	1.890		
	Ebn at Noon		148	222	252	259	250	255	228	221	232	212	140	97		
	Edh at Noon		12	23	33	41	43	39	48	42	27	18	11	7		
(44) All-Sky Solar Radiation	RadAvg		113	432	992	1601	1662	1890	1727	1328	813	359	160	53		
	RadStd		6	15	36	60	128	116	101	98	89	37	13	2		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+1.66	N/A	N/A	N/A	N/A	-551	-590	N/A	N/A
(47) Regional (7 neighbors)		+1.28	+2.59	+2.36	N/A	N/A	-425	-459	N/A	N/A

Nomenclature: See separate page